

Week 4 – Practice Session

September 28 – October 2, 2026

Problems Using Trigonometric Functions

1. One of the angles of a right triangle is 17° , and its hypotenuse is 23 cm long. How long are the legs?

(6.7 cm, 22 cm)

2. One of the angles of a right triangle is 31° , and the leg opposite this angle is 4.9 cm long. How long is the other leg?

(8.15 cm)

3. One of the angles of a right triangle is 27° , and the leg adjacent to this angle is 3.8 cm long. How long is the altitude to the hypotenuse?

(1.725 cm)

4. The sides of a rectangle are 46 mm and 79 mm long. What angle do the diagonals of the rectangle form with each other?

(60.4°)

5. A 26 cm leg of a right triangle subtends an angle of 32° . What is the radius of the circumcircle of the triangle?

(15.3 cm)

6. The altitude to the hypotenuse of a right triangle is 5 cm long. The altitude divides the hypotenuse into a segment of length 12 cm. Find the side lengths and angles of the triangle.

$\left(\frac{65}{12} \text{ cm}, 13 \text{ cm}, \frac{169}{12} \text{ cm}, 22.6^\circ, 67.4^\circ\right)$

7. An isosceles triangle has a base of 14 cm and an altitude to the base of 12 cm. What are the angles of the triangle?

(59.74° , 59.74° , 60.52°)

Problems Using the Sine Rule and the Cosine Rule

8. One of the angles of a triangle is 94° , and the side opposite this angle is 25 cm long. Another side of the triangle is 10 cm long. Find the length of the missing side and the measures of the angles.

(22.2 cm, 62.5° , 23.5°)

9. Two sides of a triangle have lengths 53 mm and 71 mm. How long is the third side if the angle between the two given sides is 147° ?

(119 mm)

10. In triangle ABC , side AB is 17 cm long. The angle at vertex A is 78° , and the angle at vertex B is 36° . Calculate the lengths of the other two sides of the triangle.

(10.9 cm, 18.2 cm)

11. A parallelogram has side lengths 10 cm and 16 cm, and one of its angles measures 100° . What are the lengths of the diagonals of the parallelogram?

(20.3 cm, 17.3 cm)

12. The sum of the lengths of two sides of a triangle is 20 cm. The angles opposite these two sides measure 67° and 48° . What are the side lengths of the triangle?

(8.3 cm, 11.7 cm)

13. The side lengths of a triangle are 6 m, 8 m, and 13 m. What are the angles of the triangle?

(18.7° , 25.3° , 136°)